Breda University of Applied Sciences | Academy of Games and Media | Data Science & AI

«<Title of Your Thesis Project»>

Bachelor's Thesis

«<Your Name»>, «<Student Number»>, «<your.email@buas.nl»>
Supervisors: «<First Supervisor»> & «<Second Supervisor»>

Abstract: «<Write your abstract here. Maximum 250 words. The abstract serves as the "teaser" of your thesis. On the basis of the abstract, readers decide whether the rest of the thesis is worth reading. In many databases, the abstract is the only freely available part.

The abstract MUST include: (1) the research problem and its context, (2) your approach or method, (3) the key findings and quantitative results, and (4) the main conclusions. It should provide a complete picture of what was done and what can be concluded from it.

The abstract must be readable as a stand-alone text: do not refer to sections, figures, or tables. Write the abstract LAST, once your results and conclusions are clear.»>

1 Introduction

Purpose: The introduction and conclusion together should provide a *complete and comprehensible* picture of your thesis. This is intended for time-poor readers whose interest has been raised by the abstract but who do not have time to read the full thesis.

Your introduction must:

- Set out the **context** of the work: why it is relevant and who has previously worked on the topic.
- State the **specific research question**, and explain how it contributes to a better understanding of the subject.
- Provide a **summary of the techniques and analysis methods** used (details are given in the Methodology section).
- **Convince the reader** that your work is useful and interesting, that you understand the subject well, and that you have chosen an appropriate way to investigate it.

Before you start writing, answer these four questions:

1. **What is the aim?** Are you reporting on a project and its results? Presenting a new idea? Proposing a solution? For a BSc thesis, the aim is usually to report on a research project and present your findings.
2. **Who is your audience?** Your supervisor, industry readers, and fellow researchers are the primary audience. They are knowledgeable but critical. Do not overexplain basic concepts, but substantiate every claim.
3. **What is the format?** A BSc thesis is typically 10–15 pages (excluding references and appendices). This is closer to a conference paper than to a journal article, so scope your depth and level of detail accordingly.
4. **What is your contribution?** State your specific contribution clearly: what message are you trying to convey? What can you assume is already known, and what background do you need to provide?

Tip: Prepare a detailed table of contents before you begin writing chapters and

sections, and add a brief description of each. This helps prevent structural problems later.

«<Write your introduction here. Start by establishing the broader context of your work, then narrow the focus to your specific research question. Provide a brief overview of your approach and convince the reader that your work is both necessary and well motivated Brown et al. (2020).»>

2 Related Work

Purpose: Position your research within the existing body of knowledge. This section is *not* a literature dump. It should present a structured argument for why your research is necessary. Every paper you cite should help build that argument.

What to include:

- Prior approaches to your problem or similar problems.
- Key results and findings from the literature.
- Theoretical foundations your work builds upon.
- Gaps in the literature that your work addresses.

How to structure:

- Group the literature by **theme or method**, not chronologically.
- **Relate each cited work** to your own research question. Do not simply summarise it; explain why it matters.
- Move from the **broader context to the specific gap** that your work fills.
- End by clearly stating how your work **extends or differs** from prior research.

Pitfalls:

- Listing papers without connecting them to your work.
- Summarising the literature without critical analysis or comparison.
- Writing literature overviews that summarise a large amount of prior work without clearly relating it to the author's own study.
- Relying only on textbooks or secondary sources.
- Overlooking recent publications (last 1–2 years).

Test: After reading this section, the reader should be able to explain (a) what has already been tried, (b) what worked and what did not, and (c) exactly which gap your thesis fills.

«<Write your related work here. Group the literature thematically, relate each reference to your research question, and clearly articulate the gap your work addresses Dawkins (1976); Brown et al. (2020).»>

3 Methodology

Golden rule: The description must focus on **reproducibility**. A reader should be able to replicate your work on the basis of this section alone. For a computer program, sufficient detail must be provided to allow the implementation to be repeated. For an experiment, sufficient detail must be given to allow the procedure to be reproduced accurately.

Three pillars:

1. **Describe what, not how**: explain the system above the source-code level. Describe what each component does, rather than how it was programmed. A good approach is to first describe how the system works as a whole, and then discuss the individual components. Substantiate the description with formulae and pseudo code.
2. **Justify your choices**: in many cases, aspects of your system can be implemented in different ways. Explain why you selected a particular method *and why you rejected the alternatives*. Place these choices in the broader context of your research question. Describing the system involves more than simply stating what you built.
3. **Include your analysis plan**: the method section should also describe the tech-

Table 1: System parameters.

Parameter	Value
Learning rate	«<0.001»>
Batch size	«<32»>
Hidden dimensions	«<256»>
Number of epochs	«<50»>
Dropout rate	«<0.3»>
Optimizer	«<Adam»>
«<...»>	«<...»>

Input: «<Input description»>
Output: «<Output description»>
«<Replace with your algorithm in pseudo code»>;
for «<*each item in dataset*»> **do**
  «<Process item»>;
  if «<*condition*»> **then**
    «<Update model»>;
  end
end
return «<*result*»>;

Algorithm 1: «<Name of your algorithm»>

niques used to *analyse* the results. Define the evaluation metrics, describe how raw data were collected, averaged, and processed, and name the statistical tests used while justifying their validity.
Important: Present all parameters in a central table (see Table 1). The reader should never have to search for a value buried in the prose.

«<Describe your methodology here. Start with a general overview, then describe the individual components. Justify every design decision and include your full analysis plan Devlin et al. (2019).»>

4 Results

Purpose: Present the essence of your message: what you discovered and how your findings shed light on the research question. Two things are especially important:

Table 2: «<Description of your results table.»>

Model	Precision	Recall	F1
Baseline	«<0.72»>	«<0.68»>	«<0.70»>
Your model	«<0.85»>	«<0.81»>	«<0.83»>

selection and **presentation**.
Selection: Present results that answer your research question. If a result does not strengthen your argument, you may leave it out.
Presentation format:
- **Line graphs** for time series or continuous relationships.
- **Bar charts** for categorical comparisons.
- **Tables** when precise numerical values matter.
- **Prose** when qualitative interpretation is the main point.

Statistical significance:
- Graphs: use error bars, box plots, or confidence intervals to show variability.
- Tables: include standard deviations, confidence intervals, and p-values.
- Text: name the statistical tests (t-test, ANOVA, Wilcoxon, etc.) and justify their validity.

Caution: Do not confuse significance with relevance. A statistically significant result can still be practically irrelevant.

«<Present your results here. Select only findings that are relevant to your research question and present them in the most appropriate format. Always include measures of statistical significance.»>

5 Discussion

Purpose: Describe the implications of your findings in the context of your research question. Guide the reader through the results by explaining how each finding relates to the research question and to prior work in the field. This discussion should build the argument for a short and, ideally, powerful final conclusion.

What to cover:

- **Interpretation:** explain how each result relates to your research question and guide the reader through that interpretation.
- **Comparison with related work:** do your findings align with, contradict, or extend prior results? Link back to Section 2.
- **Broader context:** place your conclusions in the wider academic context of the field.
- **Practical implications:** what do your findings mean for practitioners?
- **Future research:** suggest concrete future research directions that follow from your findings.

Limitations & pitfalls:

- Acknowledge limitations honestly; reviewers will identify them anyway.
- Discuss threats to validity (internal, external, construct).
- Do not assume that adding complexity is the default solution; simplification may sometimes be more effective.
- Do not confuse statistical significance with practical relevance.
- Future work should follow from your analysis, not read like a wish list.

«<Interpret your results here. Explain what they mean for the research question, compare them with related work, acknowledge limitations honestly, and suggest concrete directions for future research Vaswani et al. (2017).»>

6 Conclusion

Purpose: The introduction and conclusion together should form a *mini-article* that can be read on its own. Include all of your key findings in this section. There is no need to repeat graphs and tables, but it is essential to summarise your findings in words.

Must include:

- **Summarize key findings in words**: the reader should understand your results without needing to read the full thesis.
- **Directly answer the research question** introduced earlier. This is the single most important paragraph.
- **Briefly restate your approach** so that the conclusion can stand on its own.
- **Connect back to related work**: does your study confirm, extend, or contradict prior research?
- **Be concise**: state each conclusion clearly and separately.

Self-test: Give someone only your Introduction and Conclusion. If they can explain what you did, why you did it, what you found, and what it means, then your thesis structure is working well.

«<Write your conclusion here. Summarize your findings, directly answer your research question, and connect your conclusions back to the broader context established in the introduction.»>

DO:

- Use impersonal constructions or passive voice where appropriate.
- "This study aimed to test…" rather than "I wanted to see…"
- Substantiate every claim with references or data.
- Build paragraphs: topic sentence → detail → transition.
- Ask a peer to read and critique your draft.
- Read published papers in your field to learn the conventions.

DON'T:

- Overuse first-person pronouns. Avoid "we" in a single-author thesis.
- Make vague generalizations such as "Many researchers believe…". Cite them or remove them.
- Use overly formal language that reduces readability.
- Include material that does not serve your main message.
- Use Wikipedia as a primary source. Use it only to locate proper sources.

Detailed guidance below:

The academic style. Academic writing generally avoids first- and second-person pronouns (I, you, we). Use impersonal constructions or passive voice where appropriate. Write "this study aimed to test…" rather than "I wanted to see…". In a single-author thesis, "we" is usually inappropriate rather than helpful.

Paragraph construction. A paragraph should focus on one main subject, or link two closely related ones. The first sentence introduces the topic; the final sentence provides a transition to the next paragraph; the middle sentences provide the supporting detail. If you cannot answer "Is this relevant to my message?" positively, leave it out. If you cannot answer "Will my readers understand this?" positively, explain it in more detail.

Substantiate claims. Vague generalizations such as "The interest in X is rapidly increasing" or "Many scientists believe that…" should be avoided unless they are properly substantiated with references.

Get feedback. Writers are often the least critical readers of their own work. Ask someone else to read your text and take their comments seriously. If *you* already think something is weak, it is very likely to be weak.

Numbers and notation. Use modified American notation: points for decimals (1.5, not 1,5). Scientific notation for large numbers: 6.022×10^{23}. Use spaces to group digits: 4 294 967 296. Do not allow line breaks in the middle of formulae or between a number and its unit (e.g., 6 cm).

Formulae. Use standard mathematical notation. Explain all symbols clearly. Formulae should be left-aligned and their numbers right-aligned (e.g., Equation 4.1). Use the same font as in the main text.

Italics. Use italics for words from other languages (except commonplace terms), Latin abbreviations, new terminology, emphasis, and short quotations. Never underline.

Dashes. Hyphen (-): joins words or separates syllables. En-dash (–): minus sign, number ranges (-1, 2–5). Em-dash (—): pause or parenthetical.

Quotation marks. Use proper opening and closing marks: 'correct' and "also correct". Not 'wrong' or "wrong again".

Spelling. Use a spelling checker. Check that accents and special characters are correct. Verify hyphenation, as LaTeX may occasionally make mistakes.

Read widely. One of the most useful ways to improve your writing is to read many academic articles and observe what works and what does not. A good article will always remain a good article, and a bad one will remain a bad one. Reading both helps you understand why certain writing choices are effective.

Delete this entire box before submission.

References

Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901.

Dawkins, R. (1976). *The Selfish Gene*. Oxford University Press, Oxford.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, pages 4171–4186.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30.

A «<Title of First Appendix»>

«<Include supplementary material here: extended tables, additional figures, code listings, detailed derivations, or other material that did not fit in the main text but supports reproducibility. If you have many appendices, provide a list at the beginning.»>